Differential Equations — Cheat Sheet & Reference

Classification, solution methods, and the general terminology of the subject

Part I: classify & solve (below) · Part II: terminology (overleaf).

1. How to Classify Any Equation

ODE vs. PDE

ODE one independent variable; ordinary derivatives only.
PDE two or more independent variables; partial derivatives appear, e.g. $u_t = k\,u_{xx}$.

Order & Degree

Order the order of the *highest* derivative present.
Degree power of the highest-order derivative once the equation is written free of radicals/fractions in its derivatives (only defined when such a polynomial form exists).
$(1+(y')^2) = (y'')^{2/3} \Rightarrow$ order 2, degree 3.

Linear vs. Nonlinear

Linear y and its derivatives appear to the first power, never multiplied together, with coefficients in x only:

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x).$$

Nonlinear anything else: yy', $(y')^2$, $\sin y$, e^y, …

Autonomous?

Autonomous the independent variable is absent: $y' = f(y)$.
Non-autonomous x (or t) appears explicitly.

Side conditions

IVP all conditions at one point: $y(x_0), y'(x_0), \ldots$
BVP conditions given at two or more points.

"Homogeneous" has *two* meanings

(a) Linear sense the forcing term is zero, $g(x) = 0$; otherwise *nonhomogeneous*.
(b) First-order sense $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$, i.e. $M\,dx + N\,dy = 0$ with M, N homogeneous of the *same* degree.
Always check which sense the context intends.

2. First-Order Solvable Types

Separable

Form $\frac{dy}{dx} = f(x)\,g(y)$.
Method divide & integrate:

$$\int \frac{dy}{g(y)} = \int f(x)\,dx + C.$$

Linear (1st order)

Form $\frac{dy}{dx} + P(x)\,y = Q(x)$.
Method integrating factor $\mu = e^{\int P\,dx}$:

$$(\mu y)' = \mu Q \;\Rightarrow\; y = \frac{1}{\mu}\left(\int \mu Q\,dx + C\right).$$

Exact

Form $M(x,y)\,dx + N(x,y)\,dy = 0$ with

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Method find F with $F_x = M$, $F_y = N$; solution $F(x,y) = C$. If the test fails, seek an integrating factor.

Homogeneous (substitution)

Form $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.
Method let $v = \frac{y}{x}$, so $y = vx$, $y' = v + xv'$; the result is separable in v and x.

Bernoulli

Form $\frac{dy}{dx} + P(x)\,y = Q(x)\,y^n$, $n \neq 0, 1$.
Method substitute $v = y^{1-n}$ to obtain a linear equation in v.

Riccati (bonus)

Form $\frac{dy}{dx} = q_0(x) + q_1(x)y + q_2(x)y^2$.
Method given one solution y_1, set $y = y_1 + \frac{1}{v}$ to linearize.

3. Linear Equations of Higher Order

Constant coefficients (homogeneous)

Form $ay'' + by' + cy = 0$.
Method solve the characteristic equation $ar^2 + br + c = 0$:

- real $r_1 \neq r_2$: $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$
- repeated r: $y = (C_1 + C_2 x)e^{rx}$
- complex $\alpha \pm i\beta$:
  $y = e^{\alpha x}(C_1 \cos\beta x + C_2 \sin\beta x)$

Nonhomogeneous

General solution $y = y_h + y_p$.
Undetermined coeff. when g is a polynomial, e^{kx}, sin / cos, or their products: guess y_p of like form.
Variation of parameters for any g:

$$y_p = -y_1 \int \frac{y_2\,g}{W}\,dx + y_2 \int \frac{y_1\,g}{W}\,dx,$$

with Wronskian $W = y_1 y_2' - y_1' y_2$.

Cauchy–Euler

Form $ax^2y'' + bxy' + cy = 0$.
Method try $y = x^m$, giving $am(m-1) + bm + c = 0$; solve for m (cases mirror those above).

4. First-Order Decision Flow

Work top to bottom; stop at first match

1. Can you write $\frac{dy}{dx} = f(x)g(y)$? → **Separable**
2. Form $y' + P(x)y = Q(x)$? → **Linear** (int. factor)
3. $M\,dx + N\,dy = 0$ with $M_y = N_x$? → **Exact**
4. Depends only on y/x? → **Homogeneous**
5. $y' + Py = Qy^n$? → **Bernoulli**
6. Else: try an integrating factor, a clever substitution, or solve numerically.

Linearity, order, and homogeneity are independent axes — a single equation carries one label from each.

Part II · General Terminology & Concepts

Vocabulary every DE course leans on — solutions, conditions, series methods, transforms, and qualitative behavior.

5. Solutions & Their Behavior

Kinds of solution

General the family carrying n arbitrary constants (nth-order ODE).
Particular constants fixed by initial/boundary data.
Singular not obtainable from the general family for any constants — often the envelope of the solution curves.
Trivial $y \equiv 0$ (homogeneous linear equations).
Equilibrium constant y^* with $f(y^*) = 0$; steady-state / critical point.

Curves & forms

Integral curve the graph of one solution; the slope field is tangent to it everywhere.
Explicit $y = \varphi(x)$. **Implicit** $G(x, y) = 0$.
Solution family the general solution plotted for varying C.

Existence & uniqueness

Picard–Lindelöf if f and $\partial f/\partial y$ are continuous near (x_0, y_0), the IVP $y' = f(x, y)$, $y(x_0) = y_0$ has a *unique* solution on some interval.
Lipschitz $|f(x, y_1) - f(x, y_2)| \le L\,|y_1 - y_2|$ secures uniqueness.
Well-posed exists, is unique, and depends continuously on the data.

6. Conditions & Boundary Problems

Initial vs. boundary

Initial condition (IC) y and derivatives prescribed at *one* point x_0 → IVP.
Boundary condition (BC) conditions imposed at two or more points (the endpoints) → BVP.

Types of boundary condition

On $[a, b]$ for a 2nd-order problem:

- **Dirichlet**: value fixed, $y(a) = \alpha$
- **Neumann**: slope fixed, $y'(a) = \alpha$
- **Robin (mixed)**: $\alpha y(a) + \beta y'(a) = \gamma$
- **Cauchy**: value *and* slope at one point

Homogeneous BC the prescribed value is 0.

Eigenvalue (Sturm–Liouville) problems

A linear BVP carrying a parameter λ that admits nontrivial solutions only for special *eigenvalues* λ_n; the corresponding *eigenfunctions* form the building blocks of series solutions for PDEs.

7. Series Solutions & Special Points

Classifying a point x_0

For $y'' + P(x)y' + Q(x)y = 0$ in standard form:

- **Ordinary point**: P, Q analytic at x_0 $\Rightarrow$ two independent power-series solutions exist
- **Singular point**: P or Q fails to be analytic at x_0
- **Regular singular**: singular, yet $(x-x_0)P$ and $(x-x_0)^2Q$ are analytic $\Rightarrow$ Frobenius applies
- **Irregular singular**: singular and not regular — standard series methods fail

Power-series method

At an ordinary point set

$$y = \sum_{n\ge 0} a_n (x - x_0)^n,$$

substitute, and match like powers to get a **recurrence relation** for the a_n.
Radius of convergence at least the distance from x_0 to the nearest singular point.

Frobenius method

At a regular singular point assume

$$y = (x - x_0)^r \sum_{n\ge 0} a_n (x - x_0)^n.$$

Indicial equation the lowest-power term gives $r(r-1) + p_0 r + q_0 = 0$, fixing the exponents r_1, r_2.
The spacing $r_1 - r_2$ (integer or not) determines the form of the second solution — which may include a log term.

8. Linear-Theory Toolkit

Superposition & independence

Superposition any linear combination of solutions of a *linear homogeneous* ODE is again a solution.
Linear independence no solution is a constant-coefficient combination of the others.

Wronskian & fundamental set

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}.$$

$W \ne 0$ on an interval $\Rightarrow$ the solutions are independent.
Fundamental set n independent solutions of an nth-order linear homogeneous ODE; their span is the general solution.

Building the solution

Complementary fn. y_c general solution of the homogeneous equation.
Particular integral y_p any one solution of the nonhomogeneous equation.
Full solution $y = y_c + y_p$.
Operator $L = a_n D^n + \cdots + a_0$ with $D = \frac{d}{dx}$.

9. Transforms & Numerics

Laplace transform

$$\mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t)\,dt.$$

Turns a constant-coefficient linear IVP into algebra via $\mathcal{L}\{y'\} = sY - y(0)$; solve for $Y(s)$, then invert.

Green's function

$G(x, \xi)$ solves $LG = \delta(x - \xi)$ with the problem's homogeneous BCs. Then

$$y(x) = \int G(x, \xi)\, f(\xi)\, d\xi$$

solves $Ly = f$ — a reusable "inverse" of the operator.

Numerical methods

Euler $y_{n+1} = y_n + h\,f(x_n, y_n)$.
Runge–Kutta (RK4) weighted average of four slopes per step; far more accurate.
Stiff equation solution mixes very different scales; needs implicit solvers.

10. Systems & Qualitative Behavior

Systems & fields

System vector form $\mathbf{y}' = A\mathbf{y} + \mathbf{g}(x)$; an nth-order ODE rewrites as n first-order equations.
Slope (direction) field a segment of slope $f(x, y)$ at each point; solution curves follow it.

Phase plane, equilibria & stability

Phase portrait trajectories of an autonomous system drawn in state space.
Equilibrium a point where all derivatives vanish (critical / fixed point).
Nullcline a curve on which one component's derivative is 0.
Stability *stable* / *asymptotically stable* / *unstable* as nearby trajectories stay near / converge / diverge.
Eigenvalue test for $\mathbf{y}' = A\mathbf{y}$, the eigenvalues of A classify the equilibrium: node, saddle, spiral, or center.
Linearization approximate a nonlinear system near an equilibrium via the Jacobian.